Towards Causal Disentanglement of Latent Features in Daily CBCT for Head and Neck Cancer Radiotherapy

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Abstract

This work investigates the feasibility of using causal representation learning (CRL) methods for disentangling latent features in daily cone beam computed tomography (CBCT) scans used for radiotherapy treatment of head and neck cancer. Traditional representation learning methods often fail to differentiate between causally relevant features and spurious correlations, presenting significant challenges in clinical settings where interpretability, generalizability and accuracy are important. By employing a Variational Autoencoder, β-VAE, and the CRL iVAE models, we aim to explore the potential of CRL in extracting rich features from medical data. Our dataset includes approximately 650 cases, each comprising around 33 daily CBCT scans, providing a substantial basis for our analysis. This work is a first step in our goal to improve HNC outcomes by adapting radiotherapy treatments and reducing toxicity.

1 Introduction

Head and neck cancers (HNC) encompass a group of malignancies localized in regions such as the mouth, tongue, nose, esophagus, and neck glands. Radiotherapy, a standard treatment for HNC, involves the use of high-energy radiation to kill cancer cells and is typically administered multiple times. However, this treatment has some drawbacks. First, radiotherapy can induce toxic effects such as radionecrosis. Second, it is difficult to predict at what point during the treatment the patient has maximized the beneficial effects of the radiotherapy and we should reduce the dose. These two problems are deeply intertwined as altering the dosage to reduce toxicity may hamper chances of overall survival and vice-versa. Consequently, there is a strong incentive to investigate whether we can predict a patient's overall survival early in the treatment process to inform oncologists of the possibility of reducing the radiation dosage.

Representation learning enables models to extract rich features from data for downstream tasks. Despite its strengths, traditional representation learning often lacks interpretability and can perform poorly with out-of-distribution data due to its propensity to capture spurious correlations. In contrast, causal representation learning (CRL) has gained attention for its potential to enhance interpretability and generalizability in models. However, CRL can be more demanding in terms of data requirements, often requiring interventional data or conditioning on auxiliary variables. CRL is particularly enticing for medical applications as models which are black box and not generalizable are not desirable. Indeed, CRL models which can generalize would enable their use in different clinics without the need for retraining to reduce the domain shift. Moreover, interpretable models would boost a clinician's confidence in the model's output by allowing them to verify the reasoning behind the model's decisions.

In this work, we will evaluate the ability of models to disentangle features of medical data using both representation learning and causal representation learning frameworks. In particular, we would be interested in seeing if CRL is able to extract salient features such as tumor size or shape. We

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will specifically examine the popular Variational Autoencoder (VAE) (1) and β-VAE models (2) for representation learning, and the iVAE (3) model for CRL. Our study utilizes data from approximately 650 cases, each with around 33 daily cone beam computed tomography (CBCT) scans, collected from the CHUM.

2 Related works

To our knowledge, few works exist which investigate the potential of daily CBCTs for HNC outcome prediction. However, these works have shown that this is a promising avenue of research.

One such work by Le et al. (4) proposed a deformable 3D classification pipeline incorporating a multi-branch 3D residual convolutional neural network. This model analyzes the deformation between planning CT and longitudinal CBCT along with clinical data to predict toxicity outcomes, such as reactive feeding tube placement, hospitalization, and radionecrosis. The model achieves prediction accuracies of 85.8% for radionecrosis and 75.3% for hospitalization.

Another work by Iliadou et al. (5) proposed a machine learning method using radiomic features extracted from CBCT images to predict significant volumetric alterations which may necessitate radiotherapy treatment alterations. Their model achieved an impressive predictive accuracy of 90% from just the first week of radiotherapy scans.

While these works are innovative and help to underline the potential of CBCT scans in outcome prediction, we could argue that they are tailor-made to their data to some extent and robustness might be an issue. Moreover, the interpretability of radiomic features and deformation fields is poor. Therefore, there is a real interest in finding ways to use CRL for these tasks.

3 Methods

3.1 Dataset

For this work, we utilize HNC data collected at the CHUM since 2016. Our dataset consists of 641 patients where each case will generally have 33 data points. In total, the dataset contains 20,643 CBCT volumes. CBCT is a lower-radiation alternative to conventional CT scans that utilizes a cone-shaped X-ray beam to generate three-dimensional images. The generated images are lower quality and have different units than CT scans but are of sufficient quality to ensure that the radiotherapy is targeted correctly. Refer to Figure 1 for an example of a sagittal slice of a CT and CBCT. We also have access to the radiotherapy dose planning, the planning CT, clinical features (sex, smoking, comorbidities, etc.), and the segmentation of organs-at-risk and tumors but we did not use them for this study. However, these additional data points could be highly relevant in a future study.

For our data preprocessing, we first mask the image to remove background artefacts by combining Gaussian smoothing, automatic thresholding, and morphological opening and closing. We then resample to spacing (2, 2, 2) using a cosine windowed sinc interpolation method before cropping to 128x128x128, and then normalizing and standardizing the final volume.

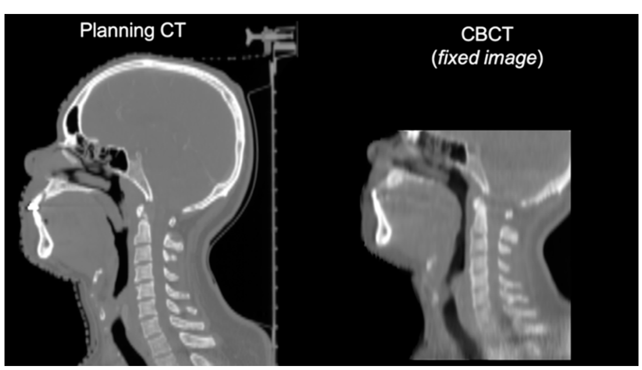

Figure 1: Example of a planning CT and a CBCT that are registered (6).

3.2 Representation Learning Methods

3.2.1 VAE and β-VAE

The Variational Autoencoder (VAE) proposed by Kingma and Welling (1) is a framework often classified as a generative model. It begins with an encoder that maps the input to a latent space and uses two separate MLPs: one to produce the mean and another for the variance of a Gaussian distribution. The reparameterization trick is then employed to generate a latent sample that incorporates stochasticity and allows for differentiability. Then, the decoder maps this latent sample back to pixel space, producing a reconstructed image of the original input. The model is optimized using the Evidence Lower Bound (ELBO), which consists of two terms: the reconstruction loss and the Kullback-Leibler (KL) divergence loss. The reconstruction loss, often the mean squared error, aims to optimize the autoencoding function of the VAE. The KL divergence acts as a regularizer, penalizing deviations of the learned latent distribution from the prior, typically the normal distribution. This objective encourages a well-structured latent space and prevents overfitting by penalizing complex models. The framework enables the model not only to reconstruct images but also to learn rich latent representations, making it suitable for representation learning. Refer to Figure 2 for an overview of the framework.

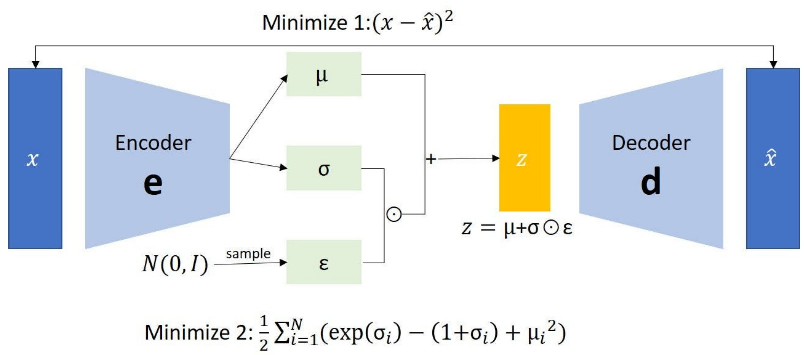

Figure 2: Overview of VAE framework (7).

A subsequent work, the β-VAE, introduced by Higgins et al. (2), uses a β hyperparameter to modulate the influence of the KL divergence term. This modification allows adjustable control over the extent to which the latent space adheres to the prior distribution. A higher β value encourages the model to prioritize learning disentangled and well-structured latent representations, albeit at the potential cost of reduced reconstruction accuracy, and vice versa.

3.2.2 iVAE

While VAE and β-VAE can be used for learning expressive features, and even disentangled features, they lack identifiability. Identifiability is the ability to uniquely recover the true underlying causal factors or mechanisms from data. To achieve identifiability with a VAE style framework, we must find a method that finds the the true prior and posterior. To address the identifiability issues, iVAE was proposed by Khemakhem et al. (3) and integrates the ideas from nonlinear ICA which allows it to not only approximate the true joint distribution over observed and latent variables but also to ensure that this approximation is identifiable up to permutation and element wise transformations. To enable this, the authors replace the Gaussian prior used for the KL divergence loss with conditional factorized priors instead. These priors are generated using an auxiliary variable which can be a class label, time index or other contextual information.

3.3 Experimental Setup

For the VAE and β-VAE, we use a convolutional encoder and decoder which are described in Table 4 and Table 5. At the bottleneck, the latent size is 512. We find that the model converges when trained for 50,000 steps with a batch size of 32 using AdamW (8) with a learning rate of 0.0001, weight decay of 0.0001 and AMSGrad (9) activated. For the data augmentations, we only use flipping along the x and y axis. For our loss function, we use the mean squared error loss for our reconstructions loss. When training our VAE model we set $\beta = 1$ and for the β-VAE we set our $\beta = 100$.

For iVAE, we need extra configurations compared to VAE. We use the treatment session number associated with the CBCT as our auxiliary variable u which we normalize based on the maximum number of radiotherapy sessions (usually 33). We then use an MLP to map u to the latent dimension which we then use for the prior in our ELBO calculation. To mimic as closely as possible the implementation proposed by the original authors, we use a different encoder to estimate the mean and the variance but retain the same convolutional architecture as our VAE for both the encoders and decoder. For the optimizer, we retain the author's default configuration: Adam (10) with a learning rate of 0.0001, weight decay of 0.01. We train for 100 epochs with a batch size of 32 which is roughly the same amount of training steps than the VAE and β-VAE. At the end of training, we observed that mean squared error validation loss had stopped decreasing.

4 Evaluation

Before diving into the evaluation, we would like to note that after trying multiple combinations of hyperparameters, architectures and data preprocessing steps, we were never able to obtain very sharp reconstructions. We believe this is due to the low quality of the original scans and to the inherent blurriness of VAE reconstructions. This may have been resolved to some extent by adding a discriminator component to the loss but optimizing Generative Adversarial Networks is notoriously tricky and we believed this was outside the scope of this project. Additionally, due to data privacy concerns, we refrained from sharing any CBCT samples in this report.

4.1 Evaluation Methodology

In the original iVAE work, they use MCC (mean correlation coefficient) between the groundtruth latents and the corresponding latents sampled from the learned posterior to evaluate how well the task was learned. In our case, we don't have access to the groundtruth latents of the CBCT. As such, we will attempt to visually interpret the reconstruction of the various models after perturbing each latent individually. Given the large latent size (512), we focus on identifying the top five latent variables that, when perturbed, result in the most significant changes in the reconstructed images within a region-of-interest (ROI) approximating the tumor's location.

To this end, we select a specific patient's CBCT at the first radiotherapy session. We take the first CBCT as we know that it will be the most similar to the planning CT which has a tumor segmentation available. We encode this volume and then perturb all the latents individually using 4 values equally spaced between -3 and 3. We then decode the perturbed latent's image and save the absolute difference at each voxel between the original reconstruction and the perturbed reconstruction. Finally given an ROI, which we defined by hand based on the CT's tumor mask, we sum the absolute difference within the region and then rank the perturbed latent's effect based on this sum.

4.2 Results

Using the methodology exposed above, we determine the top 5 latents for each model which have the most effect on the ROI. Refer to Tables 1, 2 and 3 for the latent index and the sum of difference between the original reconstruction and the perturbed reconstruction. Note that the average sum of difference in the ROI for all perturbed latents are 3.27, 2.36 and 2.77 for VAE, β-VAE and iVAE respectively.

When looking at the results, the first observation is the difference in the scale of change between the three methods. VAE engenders the most change from a single perturbed latent while iVAE engenders the least. To get a better understanding of these changes, we look at the difference maps at the center slice of the ROI. Refer to Figures 3, 4 and 5 in the Appendix for VAE, β-VAE and iVAE difference maps for each perturbed valued of their respective top 5 latent. However, based on these maps, it is difficult to make broad conclusions. In essence, it seems all three are altering the totality of the anatomical structures in the slice to some extent. We can see that the VAE has a tendency to strongly alter all structures simultaneously when a latent is perturbed which is in line with our expectations but β-VAE seems to have a similar behaviour which suggests the latents were not fully disentangled.

Of the three, iVAE seems to change the least amount of structures when a latent is perturbed but we can hardly conclude that it has isolated tumor information in these latents.

Latent Index	Perturbation Value	Sum of Difference ROI
42	3.0	386.3
42	1.8	222.9
42	-1.8	73.0
42	-3.0	87.0
85	3.0	219.2
85	1.8	165.5
85	0.6	69.3
85	-3.0	62.3
297	3.0	191.1
297	1.8	118.0
297	-1.8	103.5
297	-3.0	250.2
347	3.0	195.4
347	1.8	143.9
347	-1.8	64.6
347	-3.0	87.6
368	3.0	238.3
368	1.8	168.8
368	-1.8	67.3
368	-3.0	102.2

Table 1: Top-5 VAE perturbed latents with the largest change in the ROI of the tumor.

5 Conclusion

In this work, we explore the potential of CRL in disentangling the latent features of daily CBCT scans for patients undergoing radiotherapy for head and neck cancer. Through comparisons with VAE and β-VAE, our findings show that further refinement of the methods is required before utilizing our implementation of the CRL iVAE method for medical image analysis. Our experimental results are difficult to interpret and while we believe that the iVAE technique seems to be disentangling elements better than the other two methods, we would be hard pressed to conclude that it has disentangled the tumor features from the rest of the image as we had hoped.

In conclusion, while this particular work was not able to fully uncover a causal latent representation for daily CBCTs, our work provides a step towards leveraging these techniques. By continuing to refine these models and our implementations, we believe that their is an opportunity to enhance their performance and applicability so that we can leverage the desirable properties of CRL.

6 Future work

In the future, we believe that there are opportunities to boost the performance and evaluation of CRL techniques for this data. First, it is necessary to work on improving the reconstruction quality to help with the interpretability of the latent perturbation results. This could take the form of integrating a discriminator loss or further finetuning the architecture. Second, it would be interesting to train a linear model on top of the features produced by each VAE variant and assess them on in-distribution and out-of-distribution data for overall survival classification. Finally, it may be worth exploring the use of other auxiliary variables such as the radiotherapy dosage plan on techniques like iVAE or others like a continuous variant of BISCUIT (11).

Latent Index	**Perturbation Value**	**Sum of Difference ROI**
75	3.0	13.0
75	1.8	7.7
75	-1.8	7.7
75	-3.0	13.0
111	3.0	168.2
111	1.8	130.3
111	-1.8	70.0
111	-3.0	113.8
229	3.0	92.4
229	1.8	72.1
229	-1.8	131.5
229	-3.0	150.1
264	3.0	12.4
264	1.8	7.3
264	-1.8	7.4
264	-3.0	12.3
466	3.0	15.9
466	1.8	9.4
466	-1.8	9.1
466	-3.0	14.6

Table 2: Top-5 β-VAE perturbed latents with the largest change in the ROI of the tumor.

Latent Index	**Perturbation Value**	**Sum of Difference ROI**
206	3.0	13.7
206	1.8	8.0
206	-1.8	7.5
206	-3.0	12.3
258	3.0	9.6
258	1.8	5.9
258	-1.8	7.0
258	-3.0	12.0
332	3.0	11.3
332	1.8	6.6
332	-1.8	6.9
332	-3.0	11.7
353	3.0	12.9
353	1.8	8.0
353	-1.8	8.4
353	-3.0	14.3
454	3.0	12.5
454	1.8	7.2
454	-1.8	6.2
454	-3.0	10.3

Table 3: Top-5 iVAE perturbed latents with the largest change in the ROI of the tumor.

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Appendix A Encoder and decoder architecture description

Layer	Component	Type	Details
encode_0	conv	Conv3d	Input channels: 1, Output channels: 64, Kernel size: (3, 3, 3), Stride: (2, 2, 2), Padding: (1, 1, 1)
	adn	ADN (Activation, Dropout & Normalization)	InstanceNorm3d: 64, eps=1e-05, momentum=0.1, affine=False, track running stats=False; PReLU: num_parameters=1
	residual	Conv3d	Input channels: 1, Output channels: 64, Kernel size: (3, 3, 3), Stride: (2, 2, 2), Padding: (1, 1, 1)
encode_1	conv	Conv3d	Input channels: 64, Output channels: 64, Kernel size: (3, 3, 3), Stride: (2, 2, 2), Padding: (1, 1, 1)
	adn	ADN	Same as above
	residual	Conv3d	Same as the conv layer above
encode_2	conv	Conv3d	Same as encode_1
	adn	ADN	Same as above
	residual	Conv3d	Same as the conv layer above
encode_3	conv	Conv3d	Input channels: 64, Output channels: 128, Kernel size: (3, 3, 3), Stride: (2, 2, 2), Padding: (1, 1, 1)
	adn	ADN	InstanceNorm3d: 128, eps=1e-05, momentum=0.1, affine=False, track running stats=False; PReLU: num_parameters=1
	residual	Conv3d	Input channels: 64, Output channels: 128, Kernel size: (3, 3, 3), Stride: (2, 2, 2), Padding: (1, 1, 1)
encode_4	conv	Conv3d	Input channels: 128, Output channels: 128, Kernel size: (3, 3, 3), Stride: (2, 2, 2), Padding: (1, 1, 1)
	adn	ADN	Same as above
	residual	Conv3d	Same as the conv layer above
encode_5	conv	Conv3d	Same as encode_4
	adn	ADN	Same as above
	residual	Conv3d	Same as the conv layer above

Table 4: Detailed Architecture of the Convolutional Neural Network Encoder

Layer	Component	Type	Details
decode_0	conv	ConvTranspose3d	Input channels: 128, Output channels: 128, Kernel size: (3, 3, 3), Stride: (2, 2, 2), Padding: (1, 1, 1), Output padding: (1, 1, 1)
	adn	ADN	InstanceNorm3d: 128, eps=1e-05, momentum=0.1, affine=False, track running stats=False; PReLU: num_parameters=1
	resunit	ResidualUnit	Conv3d: 128 to 128, Kernel size: (3, 3, 3), Stride: (1, 1, 1), Padding: (1, 1, 1); Identity residual
decode_1	conv	ConvTranspose3d	Same as decode_0
	adn	ADN	Same as above
	resunit	ResidualUnit	Same as above
decode_2	conv	ConvTranspose3d	Input channels: 128, Output channels: 64, Kernel size: (3, 3, 3), Stride: (2, 2, 2), Padding: (1, 1, 1), Output padding: (1, 1, 1)
	adn	ADN	InstanceNorm3d: 64, eps=1e-05, momentum=0.1, affine=False, track running stats=False; PReLU: num_parameters=1
	resunit	ResidualUnit	Conv3d: 64 to 64, Kernel size: (3, 3, 3), Stride: (1, 1, 1), Padding: (1, 1, 1); Identity residual
decode_3	conv	ConvTranspose3d	Same as decode_2
	adn	ADN	Same as above
	resunit	ResidualUnit	Same as above
decode_4	conv	ConvTranspose3d	Same as decode_3
	adn	ADN	Same as above
	resunit	ResidualUnit	Same as above
decode_5	conv	ConvTranspose3d	Input channels: 64, Output channels: 1, Kernel size: (3, 3, 3), Stride: (2, 2, 2), Padding: (1, 1, 1), Output padding: (1, 1, 1)
	adn	ADN	InstanceNorm3d: 1, eps=1e-05, momentum=0.1, affine=False, track running stats=False; PReLU: num_parameters=1
	resunit	ResidualUnit	Conv3d: 1 to 1, Kernel size: (3, 3, 3), Stride: (1, 1, 1), Padding: (1, 1, 1); Identity residual

Table 5: Detailed Architecture of the Convolutional Neural Network Decoder

Appendix B Difference maps between the original reconstruction and the perturbed reconstructions

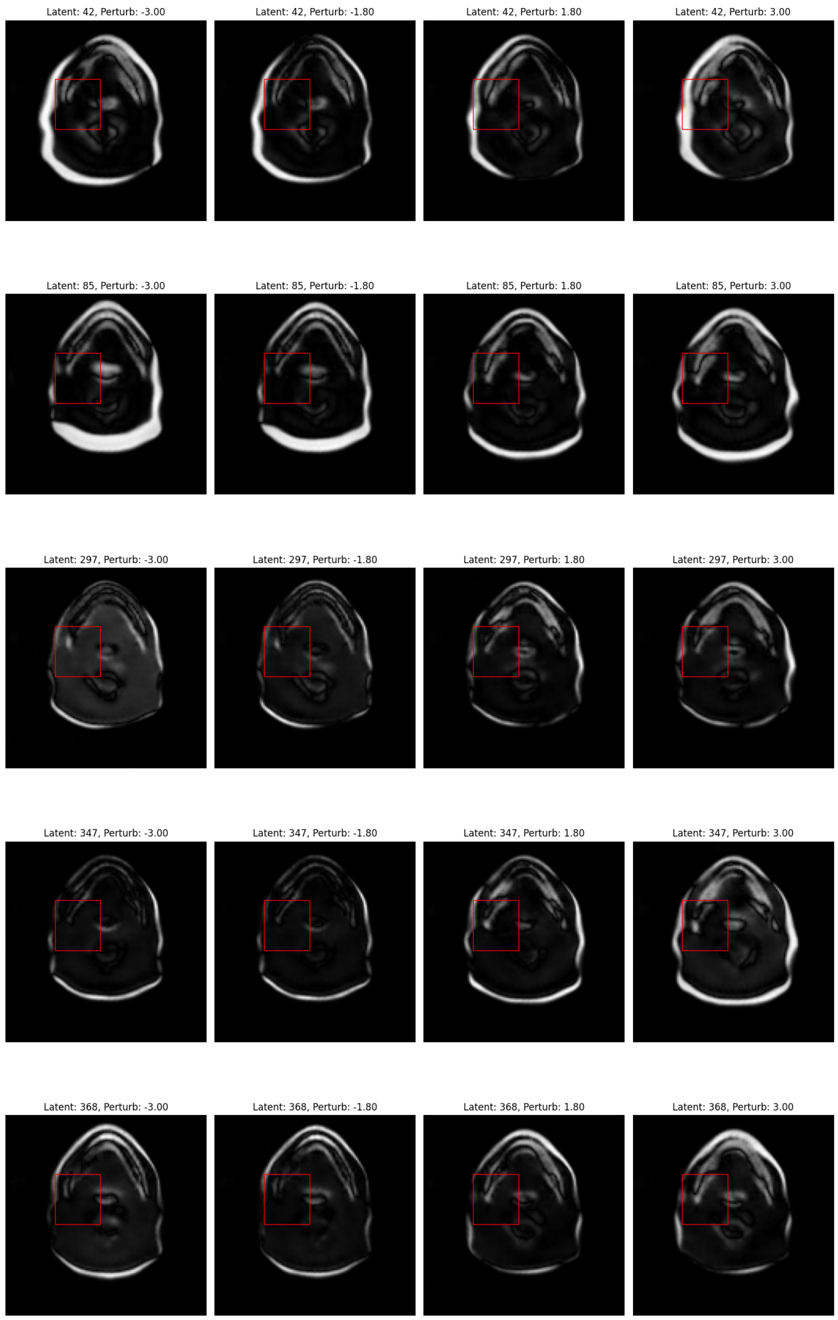

Figure 3: VAE difference maps between the original reconstruction and the perturbed latent reconstruction. The red box indicates the ROI.

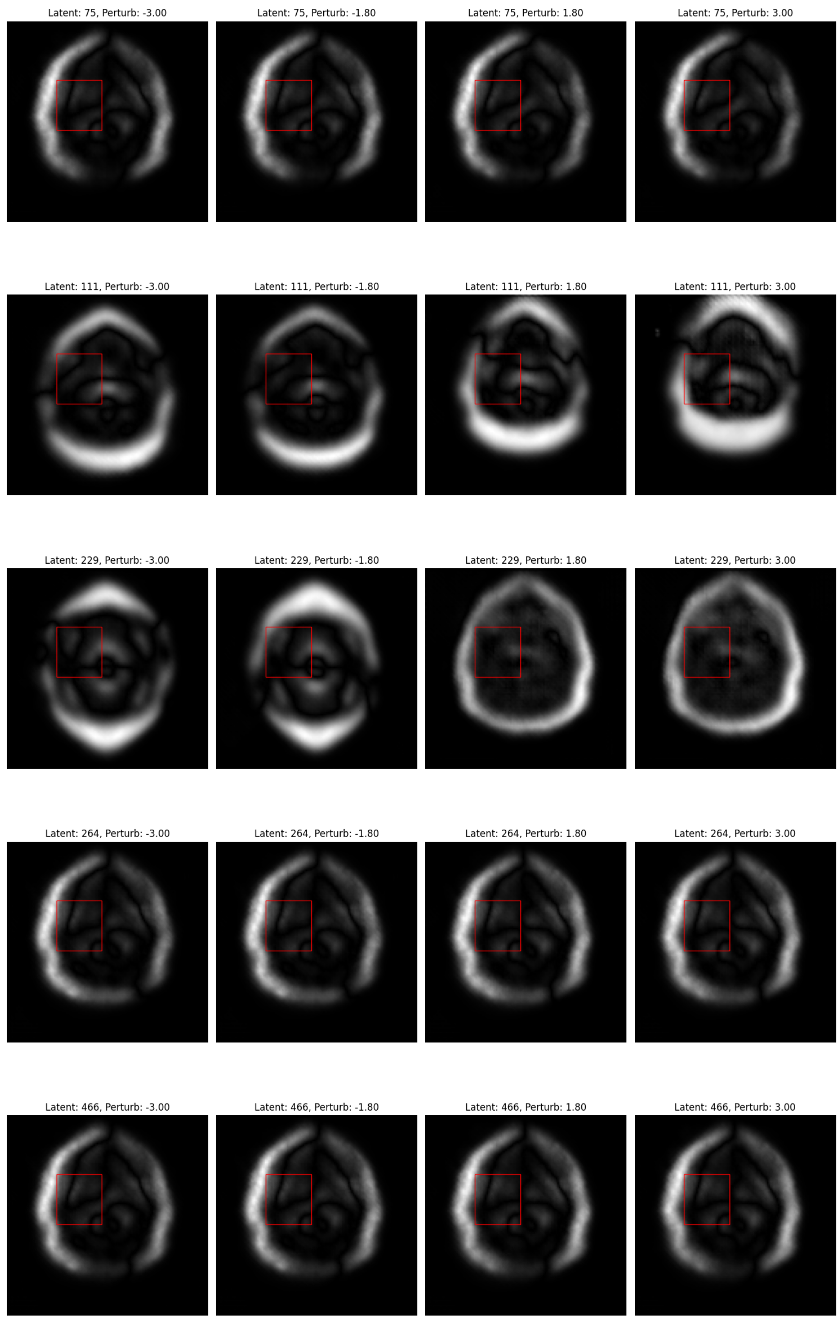

Figure 4: β-VAE difference maps between the original reconstruction and the perturbed latent reconstruction. The red box indicates the ROI.

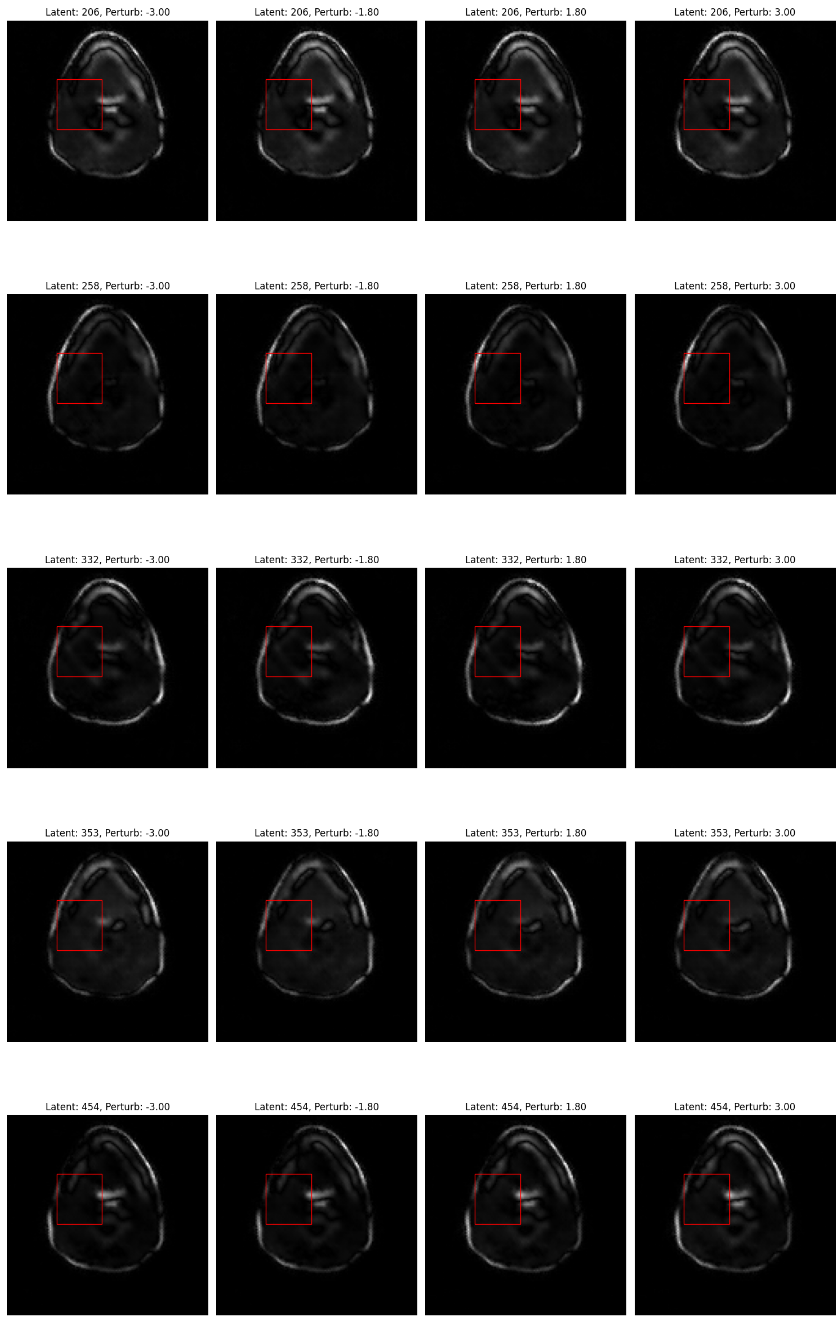

Figure 5: iVAE difference maps between the original reconstruction and the perturbed latent reconstruction. The red box indicates the ROI.